

Similarity

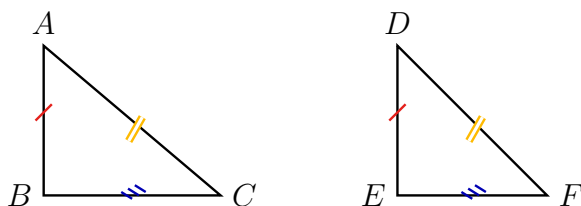
US Common Core – High School Geometry

Domain: Similarity, Right Triangles, and Trigonometry (G-SRT) | Cluster: Understand similarity in terms of similarity transformations; Prove theorems involving similarity

1 Similar Figures

Definition

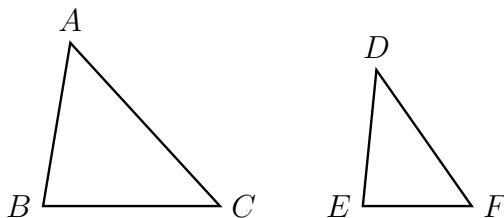
If two figures are similar, then their corresponding positions are the **same** with each other. (Compare with *congruent* figures, whose corresponding parts are exactly equal.)



Definition

If two figures are similar with each other, then their corresponding lengths are **proportional** with each other.

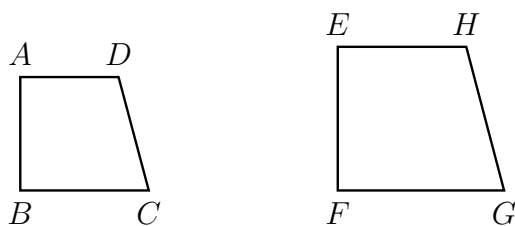
$$\frac{AB}{DE} = \frac{AC}{DF}$$



2 Similar Polygons

Definition

If two polygons $ABCD$ and $EFGH$ are similar, then their respective angles are congruent with each other, and there is a common scale factor between corresponding side lengths.



$$\angle A \cong \angle E, \quad \angle B \cong \angle F, \quad \angle C \cong \angle G, \quad \angle D \cong \angle H$$

$$\frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE} = k$$

where k is the **scale factor**.

In words

For two similar polygons $ABCD$ and $EFGH$: corresponding side lengths are proportional:

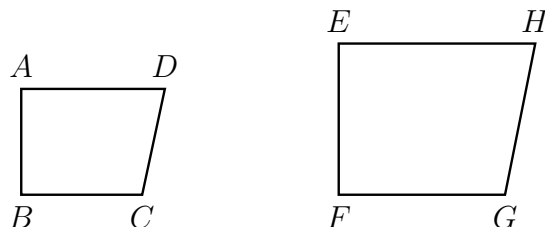
$$\frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE}$$

3 Theorem 1: Perimeters of Similar Polygons

Theorem

If two polygons $ABCD$ and $EFGH$ are similar, then

$$\frac{AB + BC + CD + DA}{EF + FG + GH + HE} = \frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE}.$$

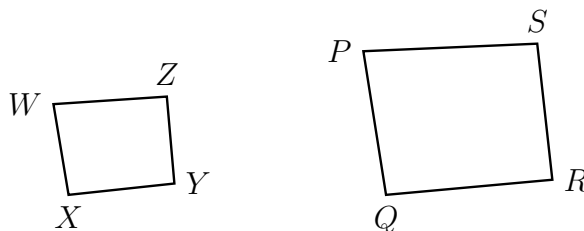


4 Theorem 2: Areas of Similar Polygons

Theorem

If two polygons $WXYZ$ and $PQRS$ are similar, then

$$\frac{\text{Area}(WXYZ)}{\text{Area}(PQRS)} = \left(\frac{WX}{PQ}\right)^2 = \left(\frac{XY}{QR}\right)^2 = \left(\frac{YZ}{RS}\right)^2 = \left(\frac{ZW}{SP}\right)^2.$$



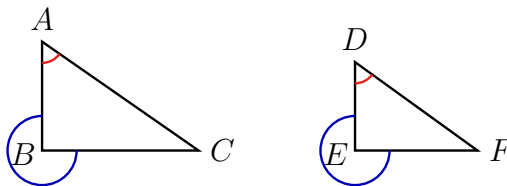
5 Triangle Similarity Theorems

*Note: The source notes state these three criteria as producing congruent triangles. Since each criterion uses proportional (not necessarily equal) sides, the mathematically correct conclusion is that the triangles are **similar**. That correction has been applied below.*

Theorem 3 (AA Similarity Theorem)

Theorem

If two angles of a triangle are congruent to the two angles of another triangle, then the two triangles are **similar** to each other.



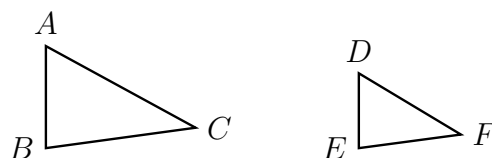
Theorem 4 (SSS Similarity Theorem)

Theorem

If all three sides of one triangle are proportional to all three sides of another triangle, then the two triangles are **similar** to each other.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

Note: The source notes wrote the first ratio as AB/DF ; corrected here to AB/DE so the vertex correspondence $A \leftrightarrow D, B \leftrightarrow E, C \leftrightarrow F$ is consistent throughout.

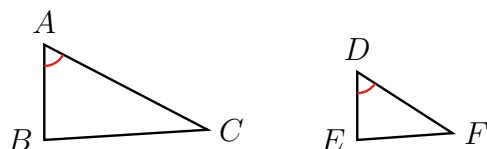


Theorem 5 (SAS Similarity Theorem)

Theorem

If one angle of a triangle is congruent to one angle of another triangle, and the two sides including that angle are proportional to the two corresponding sides of the other triangle, then the two triangles are **similar** to each other.

$$\angle A \cong \angle D \quad \text{and} \quad \frac{AB}{DE} = \frac{AC}{DF}$$

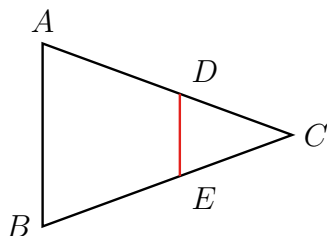


6 Triangle Proportionality Theorem

Theorem

If a segment is drawn in a triangle parallel to one of its sides, and intersects the other two sides, then it divides those two sides proportionally.

$$\overline{AB} \parallel \overline{DE} \implies \frac{AD}{DC} = \frac{BE}{EC}$$

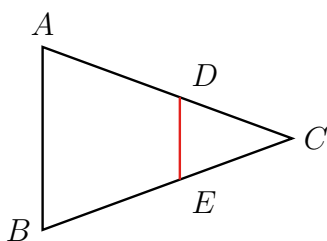


7 Converse of the Triangle Proportionality Theorem

Theorem

If two sides of a triangle are divided proportionally by two points, then the segment joining those points is parallel to the third side of the triangle.

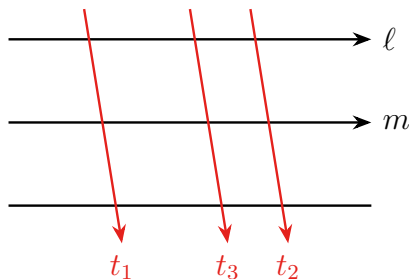
$$\frac{AD}{DC} = \frac{BE}{EC} \implies \overline{DE} \parallel \overline{AB}$$



8 Proportionality of Three Parallel Lines Cut by Transversals

Theorem

If three parallel lines are intersected by two transversal lines, then the parallel lines divide the transversal lines proportionally.



9 Triangle Angle Bisector Theorem

Theorem

If a ray bisects an angle of a triangle, then it divides the opposite side into segments proportional to the two sides forming the bisected angle.

$$\frac{AD}{DC} = \frac{BA}{BC}$$

